



REVIEW ARTICLES

Measure what matters: Single Assessment Numeric Evaluation (SANE) score as the critical measure for shoulder outcomes



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Background: Implementation of a system anchored in patient outcomes is challenged to gain widespread adoption required to demonstrate the value of care provided for shoulder conditions. This is in large part because of the administrative burden created by current tools and clinical implementation barriers that limit practical use and therefore leave most clinicians, administrators, payers, and patients without a measure of what matters most: Are patients improving? Thus, we must ask ourselves, How do we accurately and efficiently measure and report quality of care in a simple, reliable, and easily communicated manner? We propose that the Single Assessment Numeric Evaluation (SANE) score is the best solution to measure patient improvement and can be used universally for all shoulder conditions. The measure is simple, valid, reliable, and sensitive to change and has the lowest implementation barrier compared with all other outcome measures.

Methods: We synthesized the available literature (11 studies) that demonstrates strong psychometrics comparable to legacy measures across >4000 patients with a wide range of shoulder conditions.

Results: SANE scores range from 40% to 60% at baseline for most patients before treatment and range from 75% to 85% at 1 year after therapy depending on the condition, similar to legacy scores such as the American Shoulder and Elbow Surgeons score. Additionally, the SANE score shows similar baseline and post-care measures across conditions that can be used to guide clinical care. This finding shows that the observed baseline and improvement scores can provide valuable patient assessment and can be used in aggregate for quality improvement and other value-based purposes.

Conclusion: We strongly recommend the SANE score as the primary patient outcome measure for patients with all shoulder conditions, given the value of measuring every patient's progress and growing pressure to quantify patient outcomes.

Level of evidence: Narrative Review

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“You can't improve what you don't measure” is attributed to Peter Drucker to challenge businesses to “measure what matters” to provide clear, actionable goals and guide improvement activities. While health care continues to evolve from a fee-for-service system to a value-

based system, adoption of patient-reported outcome measures has failed to transform clinical practice and measure the impact of orthopedic care for shoulder conditions. This in turn leaves most clinicians without a measure of what matters—Are patients improving?—owing to the administrative and clinical implementation barriers that limit practical use. Recent large observational studies have shown that >90% of patients are missing adequate follow-up outcomes.¹³ Thus, we must ask ourselves, How do we accurately and efficiently measure and report quality of care in a reliable, simple, and easily communicated manner? There are many proven patient-reported outcome measures in shoulder surgery including the American Shoulder and Elbow Surgeons (ASES) score, Constant score (CS), and most recently, the Patient-Reported Outcomes Measurement Information System (PROMIS) score.⁵ No scoring system is perfect in its pursuit of quantifying patient outcomes as related to gain or loss function, pain, quality of life, and overall satisfaction. Patient factors such as personality, socioeconomic status, and demographic influences all contribute when designing a universally accepted patient-reported outcome score. However, the questionnaires required to serve such a diverse patient population either are administratively burdensome or require a technology investment beyond most practices' resources. This leaves orthopedic practices without a measure of patient improvement and thus provider quality to measure and guide value-based care models.

Single Assessment Numeric Evaluation score is simple and demonstrates excellent psychometrics

We propose that the Single Assessment Numeric Evaluation (SANE) score is the best solution to measure patient improvement and should be used as the universal shoulder outcome measure because it is simple, valid, reliable, and sensitive to change and has the lowest implementation barrier compared with all other outcome measures. We agree with Owens¹⁰ that clinical adoption of the SANE score will bring “sanity” to clinicians aiming to measure what matters to their patients—and their outcomes. The SANE score was introduced in the early 1980s as a simple yet powerful measure of clinical outcomes.¹⁰ The SANE score asks the patient to rate the shoulder on a scale from 0 to 100, with 100 being the patient's normal. Although singular in nature, the SANE score incorporates broad intangible differences in patient-perceived function, quality of life, and goals. This provides a measure that captures what function is important to each patient in an easy-to-implement score and communicates patient status and progress.

When considering which shoulder outcome should be used as a quality measure, we should first weigh the

importance of measuring every patient throughout his or her care journey. Although other measures may have some degree of increased accuracy and/or sensitivity, the burden of implementation on the provider, as well as the patient, results in wide variations in patient completion rates, ranging from 10% to 80%.^{7,13,14} Additionally, although most electronic medical record systems have some capability to incorporate surveys, effective implementation requires additional administrative and workflow resources to consistently obtain patient responses. In contrast to all other measures, the SANE, as a single question, is easy to implement, interpret, and evaluate at a given time point, as well as during the patient's course of care to assess treatment progress and after treatment to assess quality. As such, the SANE is an accepted outcome tool across several Centers for Medicare & Medicaid Services (CMS) Quality Payment Program measures and provides a simple bridge between day-to-day clinical medicine and overarching provider performance measurement for quality programs.⁹

When one is choosing the right tool, the assessment of patient outcome psychometrics is critical. The SANE score has been shown to be a valid and reliable way to measure shoulder status (Table I). It not only can measure change but, importantly, has the “internal control” of a patient's expectations, of his or her “normal,” built into the question. This likely explains its wide applicability from high-functioning baseball athletes to more complicated populations such as postoperative rotator cuff repair and shoulder arthroplasty (Table I).^{2,14} When comparing the SANE score across a number of conditions including shoulder instability, rotator cuff disease, and arthritis, it is comparable in all metrics to the ASES score, Western Ontario Rotator Cuff Index, Oxford score, and CS.

Specifically, studies have established the minimal clinically important difference (MCID) and substantial clinical benefit and have consistently shown that the SANE score correlates with standard outcome tools. Prior studies have shown the SANE score to be a valid and reliable patient-reported assessment for a range of common shoulder diagnoses in operative and nonoperative patients.^{7,14} The SANE score has been shown to be a consistent clinical correlate across multiple studies to other legacy outcome measures such as the ASES score ($r = 0.46-0.69$), CS ($r = 0.58$), and Rowe score ($r = 0.51-0.79$) after shoulder surgery for a variety of pathologies.^{3,4,6,16} The SANE score has been shown to be equivalent to the ASES score and CS in reporting clinically significant improvements, with MCID thresholds of 12-18 and a substantial clinical benefit threshold of 28 for rotator cuff repair and total shoulder arthroplasty at 2 years postoperatively.^{3,17} Moreover, Gowd et al⁴ showed that the SANE score is equivalent to the ASES score and CS in reporting clinically significant improvements in total shoulder arthroplasty patients. There is good evidence that the SANE score has the ability to evaluate pretreatment status, as well as differences in clinic outcomes in shoulder surgery patients.^{12,14,15} Given this

Table I SANE score psychometrics

Article title	SANE score psychometrics
Correlation of Single Assessment Numeric Evaluation (SANE) with other patient reported outcome measures (PROMs) (O'Connor and Ring, ⁸ 2019)	$r = 0.59$ (SD, 0.20) Median: 0.62 Range: 0.20–0.89
Psychometric properties of the Single Assessment Numeric Evaluation (SANE) in patients with shoulder conditions. A systematic review (Nazari et al, ⁷ 2020)	MCIDs ranging from 12% to 37% depending on statistical methodology, shoulder condition, time between measures, and initial disability
Correlation between Oxford Elbow Score and Single Assessment Numeric Evaluation: is one simple question enough? (Razaeian et al, ¹¹ 2020)	Correlation between SANE score and OES overall score: $r = 0.903$ Correlation between SANE score and OES pain score: $r = 0.804$ Correlation between SANE score and OES function score: $r = 0.847$ Correlation between SANE score and OES psychosocial score: $r = 0.885$ Correlation between SANE score and VAS score: $r = -0.641$
Establishing clinically significant outcome thresholds for the Single Assessment Numeric Evaluation 2 years following total shoulder arthroplasty (Cohn et al, ³ 2021)	MCID: 15% SCB: $\geq 81\%$ Correlation between SANE and ASES scores: $R^2 = 0.493$ Correlation between SANE and Constant scores: $R^2 = 0.586$ Correlation between SANE and Rowe scores: $r = 0.85$
Single Assessment Numeric Evaluation for instability as an alternative to the Rowe score (Ladermann et al, ⁶ 2021)	SANE AUC 70.5% for MCID: 28.8
Single Assessment Numeric Evaluation (SANE) is a reliable metric to measure clinically significant improvements following shoulder arthroplasty (Gowd et al, ⁴ 2019)	SCB for SANE score: 70.5% (AUC, 50.2)
Single Assessment Numeric Evaluation (SANE) correlates with American Shoulder and Elbow Surgeons score and Western Ontario Rotator Cuff Index in patients undergoing arthroscopic rotator cuff repair (Wickman et al, ¹⁵ 2020)	SCB for ASES score: 88% (AUC, 20.7)
Single Assessment Numeric Evaluation scores correlate positively with American Shoulder and Elbow Surgeons scores postoperatively in patients undergoing rotator cuff repair (Retzky et al, ¹² 2020)	Correlation between SANE and ASES scores: $r = 0.769$ preoperatively and $r = 0.781$ postoperatively Correlation between SANE and WORC scores: $r = 0.757$ preoperatively and $r = 0.813$ postoperatively Correlation between SANE and ASES scores in RCR patients: $r = 0.30$ preoperatively and $r = 0.86$ postoperatively Correlation between SANE and ASES scores in arthroplasty patients: $r = 0.46$ preoperatively and $r = 0.78$ postoperatively
Validity and responsiveness of the Single Alpha-Numeric Evaluation for shoulder patients (Thigpen et al, ¹⁴ 2018)	Pretreatment reliability: $ICC_{2,1} = 0.84$ (SEM, 3.8) for SANE score and $ICC_{2,1} = 0.82$ (SEM, 3.4) for ASES score SANE score MCID of 15% similar to ASES score MCID of 11%
Comparison of the Single Assessment Numeric Evaluation method and 2 shoulder rating scales. Outcomes measures after shoulder surgery (Williams et al, ¹⁶ 1999)	Correlation between SANE and Rowe scores at follow-up: $r = 0.72$ at 3 mo, $r = 0.79$ at 6 mo, $r = 0.62$ at 1 y, and $r = 0.54$ at 2 y Correlation between SANE and ASES scores at follow-up: $r = 0.66$ at 3 mo, $r = 0.69$ at 6 mo, $r = 0.64$ at 1 y, and $r = 0.64$ at 2 y
Establishing minimal important differences for the VR-12 and SANE scores in patients following treatment of rotator cuff tears (Zhou et al, ¹⁷ 2018)	MID: 4.94 for VR-12 PCS score, 5.99 for VR-12 MCS score, and 11.80 for SANE score
Can the Single Assessment Numeric Evaluation be used as a stand-alone subjective outcome instrument in patients undergoing rotator cuff repair? (Baumgarten, ¹ 2022)	Correlation between SANE and ASES scores: $r = 0.81$ ($n = 1838$) Correlation between SANE and WORC scores: $r = 0.82$ ($n = 1793$)

SANE, Single Assessment Numeric Evaluation; SD, standard deviation; MCID, minimal clinically important difference; SCB, substantial clinical benefit; ASES, American Shoulder and Elbow Surgeons; AUC, area under the curve.

The SANE score demonstrates comparable psychometrics to legacy measures across most shoulder conditions, supporting its use as a standalone patient outcome measure.

Table II SANE score clinical estimates

Article title	Patient population	Patient sample	SANE score, mean $\pm$ SD
Correlation of Single Assessment Numeric Evaluation (SANE) with other patient reported outcome measures (PROMs) (O'Connor and Ring, ⁸ 2019)	Anatomy-specific study comparing PROMs and SANE score	11 studies (6 shoulder-specific PROMs in 4 studies)	Shoulder: 62% $\pm$ 20%
Psychometric properties of the Single Assessment Numeric Evaluation (SANE) in patients with shoulder conditions. A systematic review (Nazari et al, ⁷ 2020)	Patients with shoulder conditions or disorders	9 studies (N = 2475), with samples ranging from 161 to 534 patients	Not reported
Correlation between Oxford Elbow Score and Single Assessment Numeric Evaluation: is one simple question enough? (Razaeian et al, ¹¹ 2020)	Patients who underwent consultation for elbow pathology	86 patients (107 consultations)	Before treatment: 36% $\pm$ 15% After treatment: 51% $\pm$ 26%
Establishing clinically significant outcome thresholds for the Single Assessment Numeric Evaluation 2 years following total shoulder arthroplasty (Cohn et al, ³ 2021)	TSA	153 patients	NA
Single Assessment Numeric Evaluation for instability as an alternative to the Rowe score (Ladermann et al, ⁶ 2021)	Surgical shoulders or nonoperative shoulders	253 patients	Before treatment: 45% $\pm$ 19% At final follow-up: 85% $\pm$ 15%
Single Assessment Numeric Evaluation (SANE) is a reliable metric to measure clinically significant improvements following shoulder arthroplasty (Gowd et al, ⁴ 2019)	TSA	207 patients	NA
Single Assessment Numeric Evaluation (SANE) correlates with American Shoulder and Elbow Surgeons score and Western Ontario Rotator Cuff Index in patients undergoing arthroscopic rotator cuff repair (Wickman et al, ¹⁵ 2020)	RCR	333 patients	23% improvement at 12 mo
Single Assessment Numeric Evaluation scores correlate positively with American Shoulder and Elbow Surgeons scores postoperatively in patients undergoing rotator cuff repair (Retzky et al, ¹² 2020)	RCR and arthroplasty	107 patients	RCR patients: 40% $\pm$ 19% Arthroplasty patients: 31% $\pm$ 21%

(continued on next page)

Table II SANE score clinical estimates (continued)

Article title	Patient population	Patient sample	SANE score, mean $\pm$ SD
Validity and responsiveness of the Single Alpha-Numeric Evaluation for shoulder patients (Thigpen et al, ¹⁴ 2018)	RCR, TSA, and physical therapy	77 RCR, 55 TSA, and 80 physical therapy patients	Baseline: 52% $\pm$ 16% Follow-up at 12 wk: 70% $\pm$ 16% (nonoperative patients with shoulder impingement or adhesive capsulitis)
Comparison of the Single Assessment Numeric Evaluation method and 2 shoulder rating scales. Outcomes measures after shoulder surgery (Williams et al, ¹⁶ 1999)	Various shoulder operations	209 patients	89% $\pm$ 11%
Establishing minimal important differences for the VR-12 and SANE scores in patients following treatment of rotator cuff tears (Zhou et al, ¹⁷ 2018)	Surgical and nonsurgical treatment	222 patients	Baseline: 28% (no SD provided)
Can the Single Assessment Numeric Evaluation be used as a stand-alone subjective outcome instrument in patients undergoing rotator cuff repair? (Baumgarten, ¹ 2022)	RCR	1838 patients	Preoperatively: 36 $\pm$ 21 1 y: 92 $\pm$ 11 10 y: 88 $\pm$ 16

SANE, Single Assessment Numeric Evaluation; SD, standard deviation; PROM, patient-reported outcome measure.

The SANE score shows similar clinical estimates to legacy measures across most shoulder populations, supporting its use as a standalone patient outcome measure.

wealth of evidence, the ASES Value Committee's previous recommendations on patient-reported outcomes for shoulder and elbow conditions included the SANE scores.⁵

SANE score measures clinically meaningful improvement across wide range of patients

Effective clinical implementation requires a meaningful measure to the patient and the provider. [Table I](#) summarizes the evaluated SANE scores across shoulder conditions since 1999. The studies included >4000 patients with wide variety in terms of condition, baseline function, and post-treatment follow-up. When patients undergoing shoulder surgery were evaluated, the presurgical SANE scores were consistently in the 40-60 range, whereas patients who were treated nonoperatively scored >65. Post-treatment scores at 1 year of follow-up were, on average, 78 for shoulder arthroplasty and 85 for nonoperative treatment of rotator cuff and shoulder instability disorders, rotator cuff repair and instability, mirroring reported norms for the ASES score. Additionally, the MCID ranged from 12% to 18%

for the majority of the literature across conditions, and time to follow-up was consistent with other standard scores ([Table II](#)). Larger MCIDs were associated with smaller samples, a longer time to follow-up, and an anchor-based approach, which creates artificial thresholds for improvement or success.⁷ It should be noted that these increases were consistent with variations in MCIDs across other measures. Furthermore, legacy outcome tools include pain as a discrete question or within questions on functional status and are included within the tools' overall score. Given the SANE score's strong association and robustness across conditions, it appears that the SANE score captures the impact of shoulder pain interference in a similar manner to legacy scores in a single, simple measure.

The SANE score has also shown the ability to capture pitchers' post-career shoulder function, which surprisingly is <80%.² Furthermore, reviewing our internal data from the ATI Patient Outcome Registries for Physical Therapy and Sports Medicine demonstrates that the SANE evaluation measures baseline scores and improvement in scores in a similar manner to published studies across a wide range of injured athletes and patients receiving physical therapy ([Table III](#)) (unpublished

Table III Baseline and post-treatment SANE scores and responsiveness

Patient population	Sample	SANE score, mean (SD)	SANE score responsiveness
Patients with nonspecific shoulder pain (impingement, rotator cuff, SLAP lesion, and so on) undergoing episodes of PT	1125	Baseline: 54 (22) After treatment (average of 6 wk of PT): 76 (19)	70% of patients exceeded 1 MCID (15 points) and <3% of patients displayed ceiling or floor effects at baseline or final outcome assessment
Scholastic athletes (instability, AC separation, and rotator cuff-related pain)	516	Baseline: 58 (24) Return to sport: 85 (15)	31 athletes with baseline ≥ 95 (6% ceiling) and 47 with baseline <10 (9% floor), similar to legacy outcome tools

SANE, Single Assessment Numeric Evaluation; PT, physical therapy.

Observational data from the ATI Patient Outcome Registries show adequate responsiveness and similar baseline and post-treatment SANE scores across a wide range of shoulder patients.

data, March 2024). Understanding normative SANE values on conditions provides critical information to the provider as an outcome measure to provide context and ranges for guiding and assessing quality care. The published literature and the values reported from large observational repositories show that the SANE score provides clinically meaningful insights across a wide range of shoulder patients. It appears that the majority of care-seeking patients report SANE scores in the mid 50s at the start of care and SANE scores of 76 on completion of care and 85 on return to sport.

On the basis of the ease of implementation and value of measuring every patient's progress, as well as available psychometrics, we strongly recommend the SANE score as the primary patient outcome measure for patients with shoulder conditions. With growing pressure to quantify successful outcomes, there is a role for the simplicity and commonality that the SANE score provides to patients and providers. Strong consideration should be given to making the SANE score the primary clinical measure for all shoulder patients, with other scores to be used in addition at the surgeon's discretion.

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